
VASP Workflow

User Manual

Browser GUI · CLI Agents · SLURM HPC

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June 2026 | <https://github.com/KeivanS/VASP-Flow>

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1 Introduction

VASP Workflow is a framework for setting up, running, and analysing DFT calculations with VASP. Two interfaces are provided:

- **Browser GUI** (`vasp-gui.py`) — a Flask web application: configure calculations through a form, run steps from the browser, view band-structure and DOS plots in real time.
- **CLI agents** (`vasp-agent.py` for workstations, `vasp-agent-slurm.py` for SLURM clusters) — read a plain-text `instructions.txt` file and generate all INCAR, KPOINTS, POSCAR, POTCAR, and job scripts automatically.

Both interfaces share the same `modules/` back-end.

1.1 Key features

- Functionals: PBEsol, PBE, R2SCAN, HSE06, RVV10, LDA, AM05
- Spin-orbit coupling (SOC) and collinear spin (ISPIN=2)
- Magnetic materials: correct MAGMOM per atom; spin-resolved band and DOS plots (spin-up / spin-down)
- GGA+U (Dudarev, LDAUTYPE=2) per element and orbital
- Automatic k-path detection from POSCAR lattice vectors
- Convergence testing: ENCUT and k-mesh sweeps
- DOS with orbital projections via `sumo`
- Phonons via phonopy, Wannier90 interface, DFPT Born charges
- Smart KPAR/NCORE auto-defaults scaled with MPI task count
- SLURM workflow with dependency-chained job submission

1.2 Requirements

Package	Notes
VASP 5.4+	<code>vasp_std</code> , <code>vasp_ncl</code> (SOC), optionally <code>vasp_gam</code>
POTCAR library	PAW pseudopotentials in per-element folders
Python 3.8+	Conda environment recommended
Flask	<code>pip install flask</code> — GUI only
<code>sumo</code> ≥ 2.0	<code>pip install sumo</code> — band and DOS plots
MPI	<code>mpirun</code> , <code>mpiexec</code> , or <code>srun</code>
phonopy	optional; <code>conda install -c conda-forge phonopy</code>
wannier90.x	optional; <code>conda install -c conda-forge wannier90</code>

2 Installation

```
# 1. Clone the repository
git clone https://github.com/KeivanS/VASP-Flow.git
cd VASP-Flow

# 2. Install Python dependencies
pip install flask sumo

# 3. Optional: phonons and Wannier support
conda install -c conda-forge phonopy wannier90

# 4. First-time configuration (creates site.env from template)
make setup
nano site.env      # set VASP paths, POTCAR directory, MPI command
```

2.1 site.env — platform configuration

```
PYTHON          = python3
VASP_STD        = /path/to/vasp_std
VASP_NCL        = /path/to/vasp_ncl
VASP_GAM        = /path/to/vasp_gam      # optional
VASP_POTCAR_DIR = /path/to/potpaw_PBE
MPI_LAUNCH      = mpirun -np            # or: srun
MPI_NP          = 16
WANNIER90_X     = wannier90.x
```

3 Browser GUI

3.1 Starting the GUI

```
make run      # launches on http://localhost:5001
```

On a **SLURM** cluster — run on the login node, open on your laptop via SSH port-forwarding:

```
# On your laptop:
ssh -L 5001:localhost:5001 username@cluster

# On the cluster login node:
make run

# Open http://localhost:5001 in your laptop's browser
```

3.2 Three-tab workflow

1. **Setup** — configure the project, then click *Generate Workflow*.
2. **Workflow** — run each calculation step; live VASP output streams to the browser.
3. **Results** — band-structure and DOS plots; OUTCAR analysis.

To return to a previous project use the *Select project* dropdown. → **Workflow** jumps to the run tab; **Edit & Regenerate** re-opens Setup with all settings pre-filled.

3.3 Setup tab — field guide

3.3.1 Structure and POTCAR

Field	Description
Project name	Folder created in the working directory.
POSCAR	Paste the crystal structure, or click <i>Load file</i> .
POTCAR directory	Path to your PAW library. If an element has multiple variants (Ga vs Ga_d) a chooser appears.

3.3.2 Functional

Choice	Description
PBEsol	Optimised for solids; recommended starting point.
PBE	Standard GGA; widely used.
R2SCAN	Meta-GGA; good accuracy/cost balance.
HSE06	Hybrid — accurate band gaps, significantly slower.
RVV10	van der Waals; for layered or molecular materials.
LDA / AM05	Rarely needed.

3.3.3 Spin / SOC

Choice	Effect
None	Non-magnetic.
Collinear spin (ISPIN=2)	Sets ISPIN = 2; reveals the <i>Magnetic moment / atom</i> field (Section 6.3).
SOC z/x/y	Sets LSORBIT, LNONCOLLINEAR, SAXIS; uses the vasp_nc1 binary.

3.3.4 Other flags

Field	Effect
Hexagonal BZ	Correct <i>M</i> point for hexagonal structures.
2D / slab	Sets $k_z = 1$ (Gamma-only out-of-plane).
GGA+U	Adds Hubbard <i>U</i> rows per element/orbital (Section 6.4).
Execution profile	Select <i>SLURM HPC</i> to generate sbatch scripts; edit profiles/slurm.json first.
MPI tasks	CPU cores per calculation.

3.4 Task checkboxes

Task	Directory	Notes
Structure relaxation	01_relax	IBRION=2; run first if geometry is not pre-optimised.
SCF	02_scf	Required before bands or DOS.
Band structure	03_bands	Line-mode KPOINTS; Section 7.
DOS	04_dos	Dense k-mesh; orbital projections via sumo.
DFPT	06_dfpt	Born charges + dielectric (IBRION=8).
Phonons	07_phonons	Phonopy finite-displacement; DFPT needed for NAC.
Wannier90	05_wannier	NSCF interface; Section 12.

Note. Normal execution order: **relaxation** → **SCF** → **bands** → **DOS**. Each step copies POSCAR/CHGCAR from the previous one automatically.

3.5 Band structure options

Leave the *k-path* field blank to trigger **automatic detection** from the POSCAR lattice vectors (Section 7). Enter a path manually to override, e.g. **G-M-K-G**.

3.6 DOS options

Add projection rows to request orbital-resolved DOS:

Element: **Mo** Orbitals: **d** Element: **S** Orbitals: **p**

For spin-polarised calculations sumo plots spin-up positive and spin-down mirrored below zero automatically.

3.7 Workflow tab

Each step shows three controls: **Run** (submit the step), **Edit files** (open INCAR/KPOINTS/-POSCAR in the browser; a **.bak** is created on first edit), and a **status badge**. **Run All Steps** chains every step and stops on failure.

3.8 Results tab

Panel	Contents
Band structure	Spin-polarised: blue = $\uparrow$, red dashed = $\downarrow$, same k-axis, $E_F = 0$. Non-magnetic: single-colour sumo plot.
Cumulative DOS	Stacked filled-area by element. Spin-polarised: up positive, down mirrored.
Projected DOS	Orbital-resolved via sumo; spin-down negated automatically.
OUTCAR analysis	Energy, Fermi level, magnetic moment, forces.

4 CLI Agents

4.1 vasp-agent.py — workstation

```
./vasp-agent.py
./vasp-agent.py -i my_inst.txt -s struct.vasp

cd ProjectName
./run_all.sh           # runs every step in sequence
./analyze.sh          # plots after all steps complete
```

4.2 vasp-agent-slurm.py — SLURM clusters

```
./vasp-agent-slurm.py           # uses profiles/slurm.json
./vasp-agent-slurm.py --profile slurm

cd ProjectName
./submit_all.sh                 # sbatch with dependency chaining
squeue -u $USER                 # monitor jobs
./analyze.sh                    # run locally after jobs finish
```

Two-phase workflow when convergence tests are requested:

```
./submit_convergence.sh         # STEP 1 -- submit convergence jobs
# review results, then:
./submit_calculations.sh        # STEP 2 -- prompts ENCUT/k-mesh, submits
```

4.3 Generated directory structure

```

ProjectName/
  instructions.txt
  POSCAR
  POTCAR                built once from $VASP_POTCAR_DIR
  01_relax/             INCAR  KPOINTS  POSCAR  POTCAR  run.sh
  02_scf/
  03_bands/
  04_dos/
  analysis/             plots generated by analyze.sh
  run_all.sh            workstation sequential runner
  submit_all.sh         SLURM job submission (dependency chained)
  analyze.sh            post-processing: plots + OUTCAR extraction

```

5 instructions.txt Format

5.1 Example

```

Project: My material name

Methods: PBEsol functional
         collinear magnetization
         GGA+U with U=4.0 on Fe-d orbitals

Tasks: structure relaxation
       SCF calculation
       band structure
       DOS (projected on Fe-d and O-p)

Convergence: Test k-points 6x6x6 to 12x12x12
             ENCUT 400-600 eV

MAGMOM: 4.0
NSW: 100
EDIFFG: -0.01
MPI: 128
NODES: 2
NTASKS_PER_NODE: 64
WALLTIME: 12:00:00

```

The parser is case-insensitive and searches the entire file, so section order is flexible.

5.2 Keyword reference

Keyword	Meaning
Project:	Project name; becomes the output folder.
MAGMOM:	Initial moment in μ_B /atom (collinear only). Written as MAGMOM = N*value . Default 2.0.
ISIF:	Relaxation type: 3=full, 2=ions only, 4=fixed volume.
NSW:	Max ionic steps. Default 100.
EDIFFG:	Force criterion in eV/Å. Default -0.01 .
ENCUT:	Manual plane-wave cutoff in eV.
NKPTS:	k-points per band segment. Default 40.
MPI:	Total MPI tasks (nodes $\times$ cores/node).
KPAR:	Global k-point parallelisation.
NCORE:	Global cores-per-orbital.
SCF_KPAR: <i>etc.</i>	Per-step KPAR: also BANDS_, DOS_, RELAX_, DFPT_, PHONONS_.
SCF_NCORE: <i>etc.</i>	Per-step NCORE; same prefixes.
NODES:	Compute nodes (SLURM).
NTASKS_PER_NODE:	MPI tasks per node (SLURM).
PARTITION:	SLURM partition.
WALLTIME:	Time limit, e.g. 12:00:00.
ACCOUNT:	SLURM account / allocation.
DFPT_EDIFF:	EDIFF for DFPT. Default $1E-8$.
PHONONS_DIM:	Supercell, e.g. 2 2 2.
PHONONS_MESH:	q-mesh, e.g. 20 20 20.
PHONONS_DISP:	Displacement in Å. Default 0.01.
PHONONS_BAND:	High-symmetry path for phonon bands.
PHONONS_NAC:	FALSE to disable NAC.
WANNIER_NUM_WANN:	Number of Wannier functions.
WANNIER_PROJ:	Projections, e.g. Ga:s;As:p .
WANNIER_EWIN:	Energy window, e.g. -5.0 to 15.0 .

6 Methods

6.1 Functionals

```

Methods: PBEsol functional      # recommended for solids
Methods: R2SCAN functional      # meta-GGA
Methods: HSE06 functional       # hybrid -- more accurate but slower
Methods: RVV10 functional       # van der Waals

```

Default ENCUT values: PBE 400 eV, PBEsol 450 eV, R2SCAN 680 eV, HSE06 400 eV, RVV10 450 eV.

6.2 Spin-orbit coupling

```

Methods: SOC with magnetization in z-direction

```

Uses `vasp_ncl`. Sets `LSORBIT`, `LNONCOLLINEAR`, and `SAXIS`.

6.3 Magnetic materials (collinear spin)


```
Methods: collinear magnetization
MAGMOM: 4.0
```

Sets `ISPIN = 2` and `MAGMOM = N * value` in every INCAR, where N is the atom count from the POSCAR.

Tip. Typical starting moments: Fe, Co $\approx 4\text{--}5 \mu_B$; Ni $\approx 2 \mu_B$; Cr, Mn $\approx 3\text{--}4 \mu_B$; rare earths $\approx 7 \mu_B$. VASP converges the self-consistent value automatically.

Antiferromagnets — after generation, edit INCAR to alternate signs:

```
MAGMOM = 4.0 -4.0 4.0 -4.0 ! alternating per sublattice
```

Spin-resolved plots produced automatically after DOS/bands run:

- *Cumulative DOS* — spin-up above the x-axis; spin-down mirrored below zero; elements colour-coded; solid = $\uparrow$, dashed = $\downarrow$.
- *Band structure* — spin-up in blue, spin-down in red dashed, same k-axis, $E_F = 0$.
- *Projected DOS* — sumo natively negates the spin-down channel in the orbital-resolved PDF.

Check the converged moment:

```
grep "magnetization_(x)" OUTCAR | tail -5
```

6.4 GGA+U

```
Methods: GGA+U with U=4.0 on Fe-d orbitals
          GGA+U with U=3.0 on Ni-d orbitals
          GGA+U with U=6.0 on Ce-f orbitals
```

Dudarev scheme (LDAUTYPE=2); $J = 0$ is set automatically. Combine freely with `collinear magnetization`.

7 K-Path for Band Structure

7.1 Auto-detection (recommended)

Omit the path in `instructions.txt` or leave the field blank in the GUI:

```
Tasks: band structure # no "along ..." -> auto-detect from
      POSCAR
```

The generator reads POSCAR lattice vectors, computes cell lengths and angles, and selects the standard path.

Type	Detection criterion	Default path
cF (FCC)	$a=b=c$, angles $\approx 60^\circ$	G-X-W-K-G-L-U-W
cI (BCC)	$a=b=c$, angles $\approx 109.5^\circ$	G-H-N-G-P-H
cP (cubic)	$a=b=c$, 90°	G-X-M-G-R-X
hP (hexagonal)	$a \approx b$, $\gamma = 120^\circ$	G-M-K-G-A-L-H-A
tP (tetragonal)	$a \approx b \neq c$, 90°	G-X-M-G-Z-R-A-Z
oP (orthorhombic)	90° , $a \neq b \neq c$	G-X-S-Y-G-Z-U-R-T-Z
rP (rhombohedral)	$a=b=c$, equal angles	G-T-F-G-L
mP (monoclinic)	$\alpha = \gamma = 90^\circ$, $\beta \neq 90^\circ$	G-Y-A-Z-G-B-D

Note. For **conventional** FCC or BCC cells (right angles, 4 or 2 atoms), auto-detection returns the cubic (cP) path. Use the **primitive** cell for an unfolded FCC/BCC band structure — the primitive FCC cell has angles $\approx 60^\circ$ and is recognised automatically.

The detected type appears in the KPOINTS comment:

```
Line-mode KPOINTS [FCC -- auto-detected]
```

7.2 Manual override

```
Tasks: band structure along G-X-W-K-G-L
```

Label	Coords	Use	
G	(0, 0, 0)	all	Γ
M	(0.5, 0.5, 0)	cubic, tet	(0.5, 0, 0) in hexagonal
K	(1/3, 1/3, 0)	hexagonal	
R	(0.5, 0.5, 0.5)	cubic	
Z	(0, 0, 0.5)	general	
A	(0, 0, 0.5)	hex, tet	
L	(0.5, 0, 0.5)	hexagonal	
H	(1/3, 1/3, 0.5)	hexagonal	
S	(0.5, 0.5, 0)	orthorhombic	
W	(0.5, 0.25, 0.75)	FCC primitive	
P	(0.25, 0.25, 0.25)	BCC primitive	
N	(0, 0, 0.5)	BCC primitive	

8 DOS Projections

All of the following forms are accepted:

```
DOS (projected on Mo-d and S-p)
DOS with projections on Ni:d and Ni:s
DOS with projection on Fe-d and O-p
```

Orbital channels: s, p, d, f. Separator is either - or :. Sumo generates an orbital-resolved PDF in analysis/.

9 Convergence Tests

```
# Range (auto-steps)
Convergence: Test k-points from 6x6x1 to 18x18x1 # 2D material
              Test k-points from 6x6x6 to 12x12x12 # 3D material
ENCUT 400-600 eV

# Explicit mesh list
Convergence: Test k-points 6x6x3, 12x12x6, 18x18x9
```

Results in 00_convergence/kpoints/ and 00_convergence/encut/.

```
./run_convergence.sh # workstation
./submit_convergence.sh # SLURM
./analyze_convergence.sh # plots: energy / pressure / forces
```

10 Parallelization

10.1 Auto-defaults

When KPAR/NCORE are not specified the agent computes from total tasks N :

Step	KPAR	NCORE	Notes
SCF / DOS / relax	$\lfloor \sqrt{N} \rfloor$	N/KPAR	
bands	1	N	Required for line-mode
DFPT	1	N	Required for IBRION=8
phonons	1	$\lfloor \sqrt{N} \rfloor$	

Note. $\text{KPAR} \times \text{NCORE}$ must divide N evenly. The agent rounds down automatically to the nearest valid value.

10.2 Manual overrides

```
MPI: 128          KPAR: 4          NCORE: 8

# Per-step (take priority over globals)
SCF_KPAR: 4       BANDS_KPAR: 1     DOS_KPAR: 8
SCF_NCORE: 8      BANDS_NCORE: 16    DOS_NCORE: 8
RELAX_KPAR: 2     DFPT_KPAR: 1      PHONONS_KPAR: 1
```

11 SLURM Cluster Settings

11.1 profiles/slurm.json

```
{
  "vasp_std": "vasp_std", "vasp_ncl": "vasp_ncl",
  "mpi_cmd": "srun",
  "modules": ["intel/2021", "impi/2021", "vasp/6.3.2"],
  "slurm": {
    "partition": "standard",
    "nodes": 2,
    "ntasks_per_node": 64,
    "time": "12:00:00",
    "account": "mygroup"
  }
}
```

11.2 Per-project overrides (in instructions.txt)

```
NODES: 4          NTASKS_PER_NODE: 32      PARTITION: gpu
WALLTIME: 48:00:00 ACCOUNT: myallocation
```

12 Wannier90

```
Tasks: Wannier90 wannierization

WANNIER_NUM_WANN: 8
WANNIER_PROJ: Ga:s;As:p
WANNIER_EWIN: -5.0 to 15.0
```

The agent generates `05_wannier/wannier90.win` with the NSCF k-mesh and projection block. After the NSCF run, execute `wannier90.x` from the `05_wannier/` directory.

13 Phonons (phonopy)

```
Tasks: phonons lattice dynamics

PHONONS_DIM: 2 2 2
PHONONS_MESH: 20 20 20
PHONONS_DISP: 0.01
PHONONS_BAND: G M K G A
PHONONS_NAC: FALSE          # disable non-analytic correction
```

Note. For the non-analytic correction (important for polar materials), include DFPT in your task list so Born effective charges are computed.

14 Worked Examples

14.1 MoS₂ monolayer with SOC

```
Project: MoS2 monolayer

Methods: PBEsol functional
         SOC with magnetization in z-direction

Tasks: structure relaxation
       SCF calculation
       band structure
       DOS (projected on Mo-d and S-p)

Convergence: Test k-points 8x8 to 20x20
             ENCUT 400-600 eV

MPI: 64
```

The generator detects hexagonal (hP) and uses G-M-K-G-A-L-H-A.

14.2 Nickel: magnetism + GGA+U

```
Project: Nickel GGA+U

Methods: PBEsol functional
         collinear magnetization
         GGA+U with U=4.0 on Ni-d orbitals

Tasks: structure relaxation
       SCF calculation
       band structure
       DOS (projected on Ni-d and Ni-s)

MAGMOM: 2.0
MPI: 128
NODES: 2
NTASKS_PER_NODE: 64
WALLTIME: 08:00:00
```

With a primitive FCC POSCAR (angles $\approx 60^\circ$) the generator detects cF and uses G-X-W-K-G-L-U-W. Band and DOS plots show both spin channels.

14.3 GaAs: phonons with NAC

```
Project: GaAs phonons

Methods: PBEsol functional

Tasks: SCF calculation
       DFPT Born charges and dielectric
       phonons lattice dynamics

PHONONS_DIM: 4 4 4
PHONONS_MESH: 20 20 20
PHONONS_BAND: G X W L G K

MPI: 64      WALLTIME: 24:00:00
```

14.4 Bi₂Se₃: topological insulator

```
Project: Bi2Se3 topological

Methods: PBEsol functional
        SOC with magnetization in z-direction

Tasks: structure relaxation
       SCF calculation
       band structure
       DOS (projected on Bi-p and Se-p)
       Wannier90 wannierization

Convergence: Test k-points from 8x8x4 to 16x16x8
             ENCUT 400-600 eV

WANNIER_NUM_WANN: 18
WANNIER_PROJ: Bi:p;Se:p
WANNIER_EWIN: -2.0 to 2.0
MPI: 128
```

15 Common OUTCAR Checks

grep "reached_required_accuracy"	OUTCAR	# converged?
grep "energy_without"	OUTCAR tail -1	# total energy
grep "E-fermi"	OUTCAR tail -1	# Fermi level
grep "magnetization(x)"	OUTCAR tail -5	# moment
grep "TOTAL-FORCE"	OUTCAR tail -20	
grep -A3 "BORN_EFFECTIVE_CHARGES"	OUTCAR head -20	

16 Troubleshooting

Symptom	Solution
POTCAR not found	Check <code>VASP_POTCAR_DIR</code> in <code>site.env</code> or GUI System Setup. Use the POTCAR chooser for unusual variants (<code>Ga_d</code> , <code>Fe_sv</code>).
<code>vasp_std</code> not found	Set the full binary path in <code>site.env</code> or GUI System Setup.
WARNING: may not have converged	Increase NSW (ionic) or NELM (electronic) in INCAR and re-run.
Wrong bands ($W = \Gamma$)	Conventional cubic cell detected; use the <i>primitive</i> cell for FCC/BCC so auto-detection picks cF/cI.
Projected DOS missing	Check <code>instructions.txt</code> : <code>projected on</code> and <code>projections on</code> are both accepted. Verify element names match POSCAR.
KPAR error	$KPAR \times NCORE$ must divide total MPI tasks. Use auto-defaults or set explicitly valid values.
GUI unreachable	Check SSH tunnel: <code>ssh -L 5001:localhost:5001 user@cluster</code> .
Project missing from dropdown	Folder must contain POSCAR + <code>settings.json</code> in the directory where <code>vasp-gui.py</code> was started.

17 Version History

- v1.0 (2026-03-23)** Initial release — GUI, workstation agent, PBEsol/R2SCAN/HSE06, SOC, GGA+U, convergence tests, Wannier90, phonopy.
- v1.1 (2026-04-05)** Per-task KPAR/NCORE smart defaults; SLURM agent with dependency-chained submission; cumulative DOS by element (no pymatgen); copy-from-relax/scf scripts.
- v1.2 (2026-06)** Collinear magnetism: MAGMOM per atom, spin-resolved band and DOS plots. Auto k-path detection (8 crystal systems). Fixed DOS projection parsing. Separate GUI and agent quick-reference sheets. Rewritten comprehensive LaTeX manual.